

IEEE NEWCAS 2026 • LECTURE • JUNE 22, 2026

Score Shaping & Pareto Optimality in Analog IC Design Automation

Why the *shape* of your score function decides whether automated transistor sizing succeeds.

 sigmoid score

 linear baseline

Danial Noori Zadeh

MacAnalog • McMaster University — Electrical & Computer Engineering

McMaster University | **Engineering**

MacAnalog — analog design automation at McMaster

Danial Noori Zadeh

Electrical & Computer Engineering, McMaster University. Building open tooling that puts **optimization formulation** — not just the solver — at the centre of analog sizing.

- SpiceXplorer: YAML-driven black-box optimization over SPICE.
- This talk: a controlled study of how the score-function shape changes what the optimizer finds.

1

One message for the next 12 minutes

*Problem **formulation** matters as much as the solver. A bounded (sigmoid) score meets every spec where the unbounded (linear) baseline silently blows the current budget — and finds **~2× lower-power** designs.*

Sizing is a constrained, multi-objective trade-off

Analog specs live on wildly different scales — and how you normalize them decides what the optimizer chases.

The setup

- Find a parameter vector x (widths, lengths, biases) that meets hard targets T , then maximizes a figure of merit.
- Objectives are application-dependent: bandwidth, gain, noise, power, phase margin, settling — all at once.

The gap

- Research over-fixates on the solver; the score function is under-studied.
- An ill-shaped objective stalls even a strong optimizer.



A 500 MHz bandwidth error and a 100% power error live on totally different scales. Normalize them wrong and one drowns the other.

Constraint satisfaction, then a single fitness

Feasibility first (drive every penalty to zero), then bank reward beyond spec — one scalar $F(x)$.

Raw penalty per metric

EXCEED

$$p = m \max(0, T - m)$$

MINIMIZE

$$p = m \max(0, m - T)$$

EXACT

$$p = |m - T|$$

m = measured metric, T = target. Zero penalty = spec met.

Dual-mode fitness $F(x)$

WHILE INFEASIBLE

$$F(x) = - \sum P_i \quad (\text{push penalties} \rightarrow 0)$$

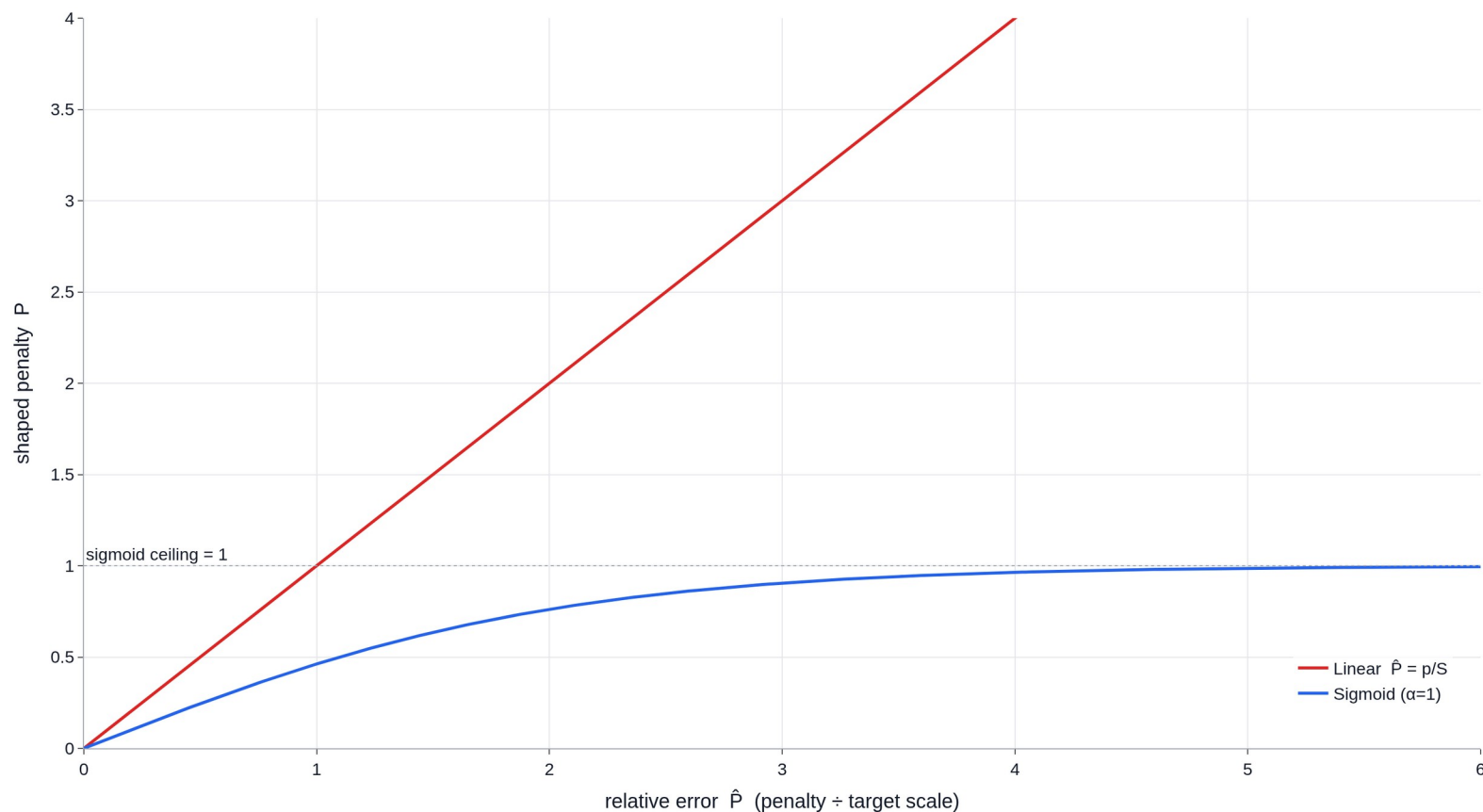
ONCE FEASIBLE

$$F(x) = + \text{reward}(x)$$

Crossing $F = 0$ means every constraint is met.

Score shaping: bound the penalty

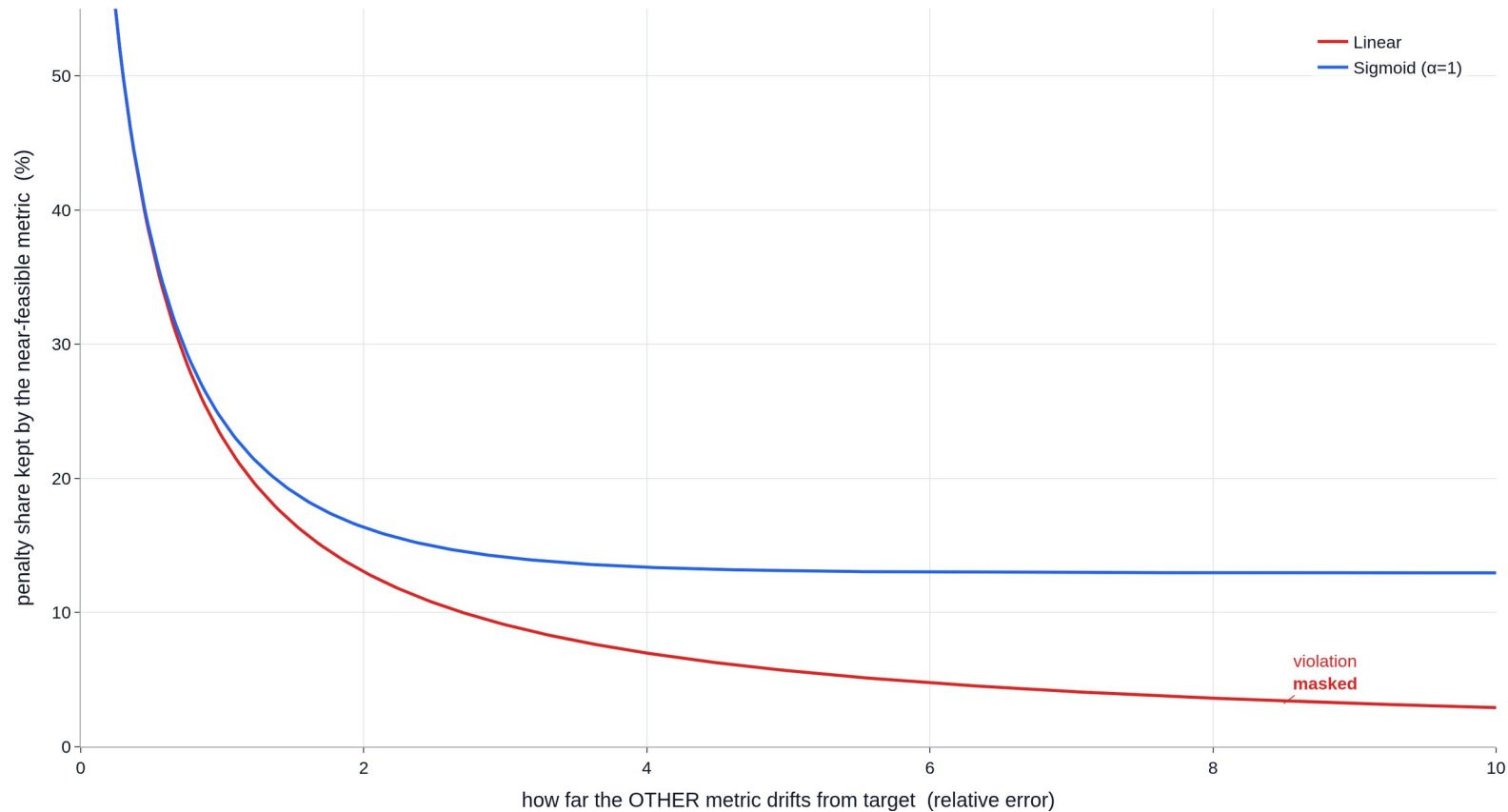
Linear (red) $\hat{P} = p/|T|$ grows without limit — one outlier dominates. Sigmoid (blue) saturates at 1 — balance beats peaks.



Linear vs. sigmoid penalty mapping · α sets the saturation rate

The masking effect

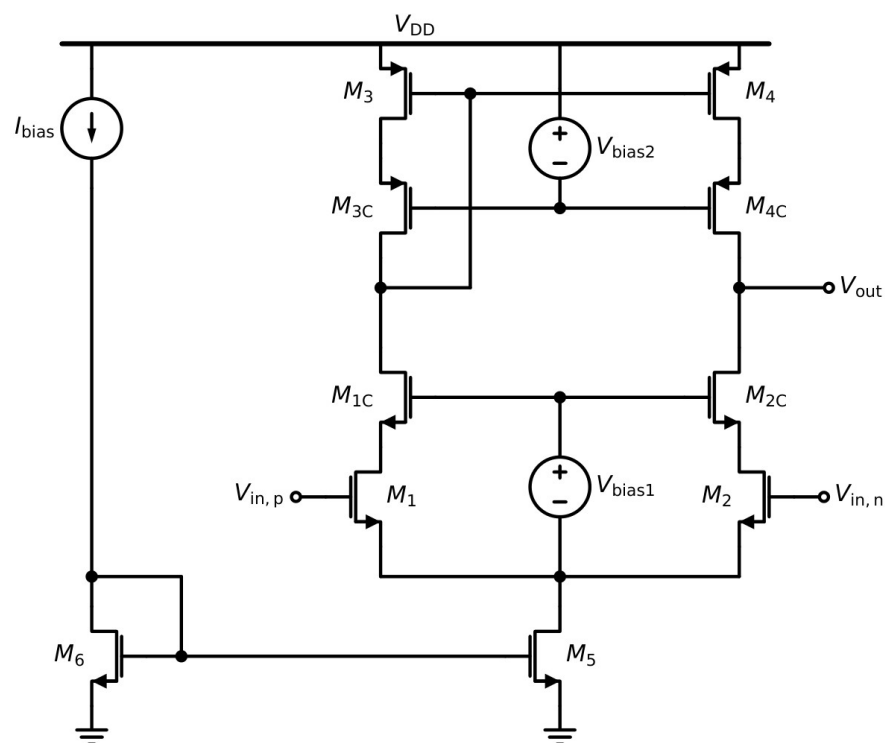
Under linear scoring, a metric just outside spec loses its share of the cost as another metric drifts — its violation is *masked*. The bounded sigmoid keeps it visible.



Penalty share kept by a near-feasible metric (rel. error 0.30) as another drifts

Telescopic cascode OTA — IHP 130 nm

14-D design space (W/L + biases). Three engines, same targets, a resource-constrained budget.



Improved telescopic cascode OTA, ideal bias (paper Fig. 2)

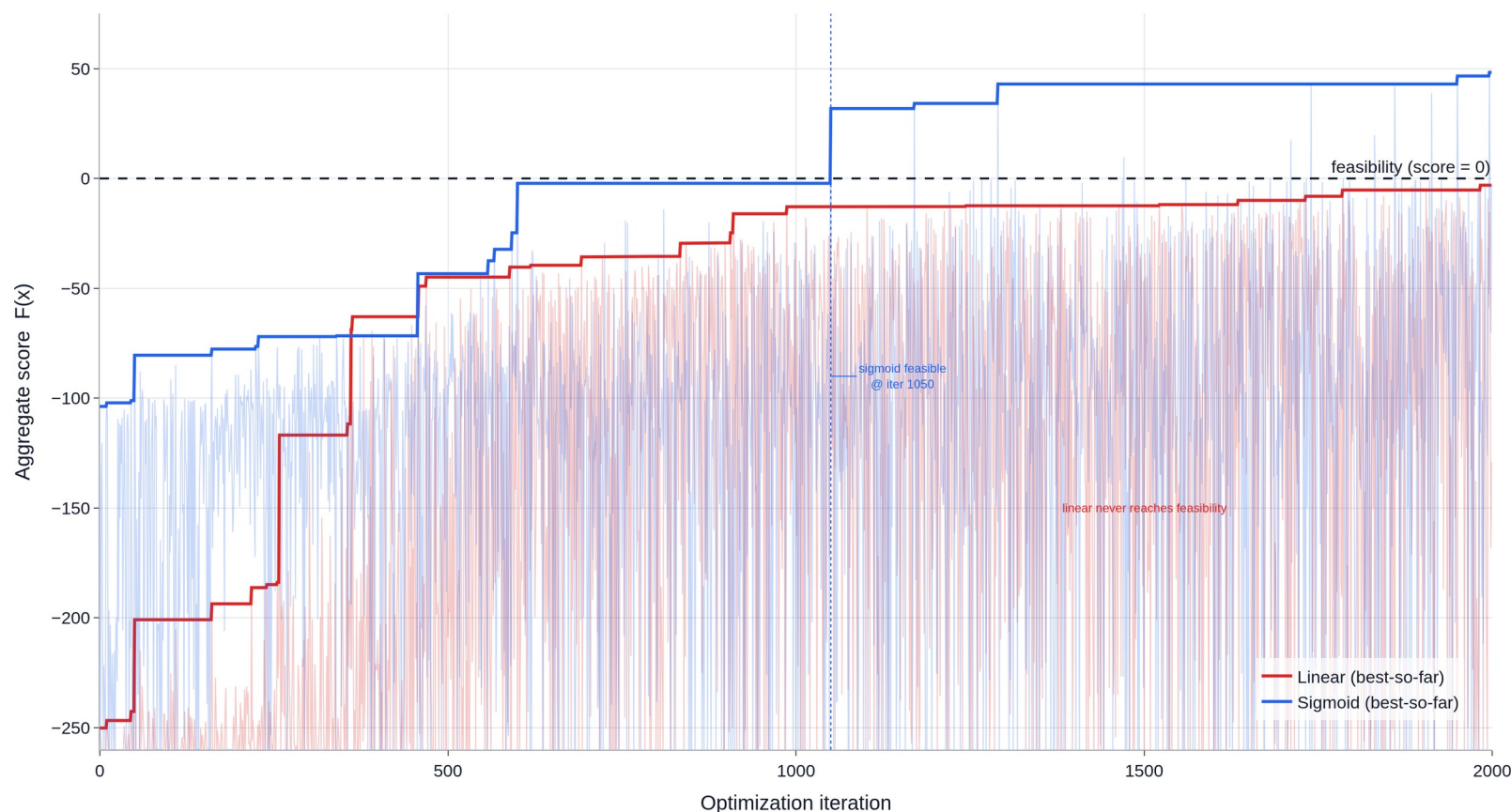
Metric	Target	Type
Unity-gain freq.	>200 MHz	max
DC gain	>40 dB	max
Input noise	<1.2 mV	min
Supply current	<25 μ A	min
Phase margin	$60 \pm 10^\circ$	exact
Settling time	<15 μ s	min

- LHS baseline 12k sims
- DE — the workhorse 2k sims
- BFGS 2k sims

V_{DD} 1.5 V · C_L 50 fF · I_{bias} 5 μ A · 14-D search space

Same solver, same budget — only the score shape changed

Sigmoid crosses into feasibility at iteration 1050 and banks reward. Linear **never** reaches a feasible design in 2,000 sims.



SIGMOID

feasible @

iter 1050

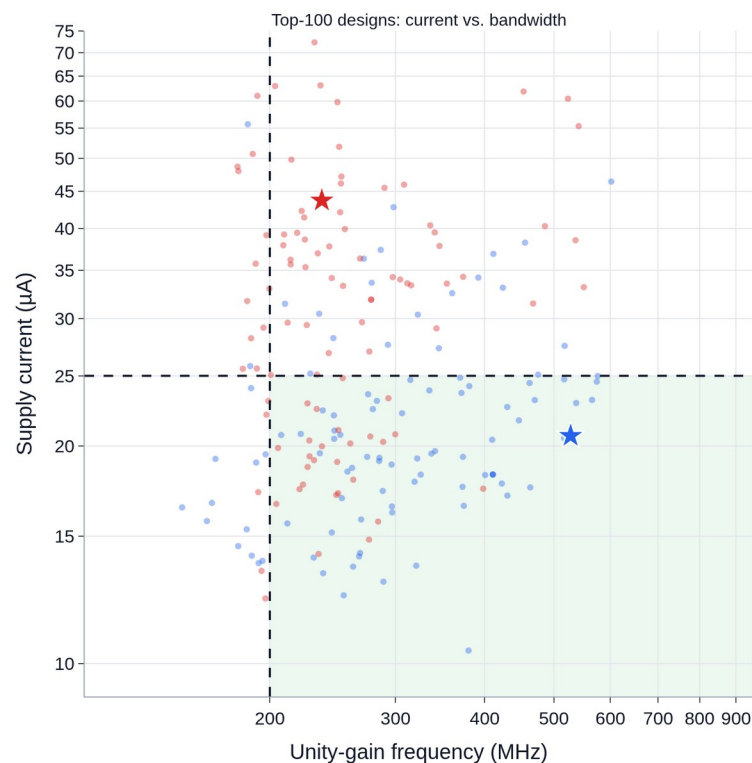
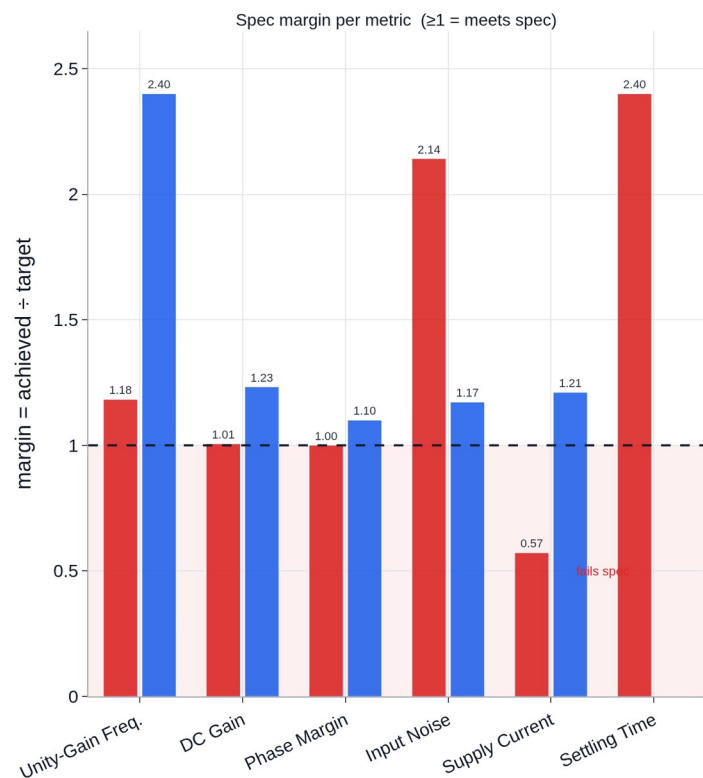
LINEAR

feasible designs

0 / 2000

Sigmoid balances every spec; linear blows the current budget

Linear's best maximizes bandwidth & gain but lands at **43.7 μA — 1.75× over the 25 μA budget**. Sigmoid meets all six specs at **20.6 μA** . The masking effect, made real.



SIGMOID BEST

20.6 μA

meets all 6 specs

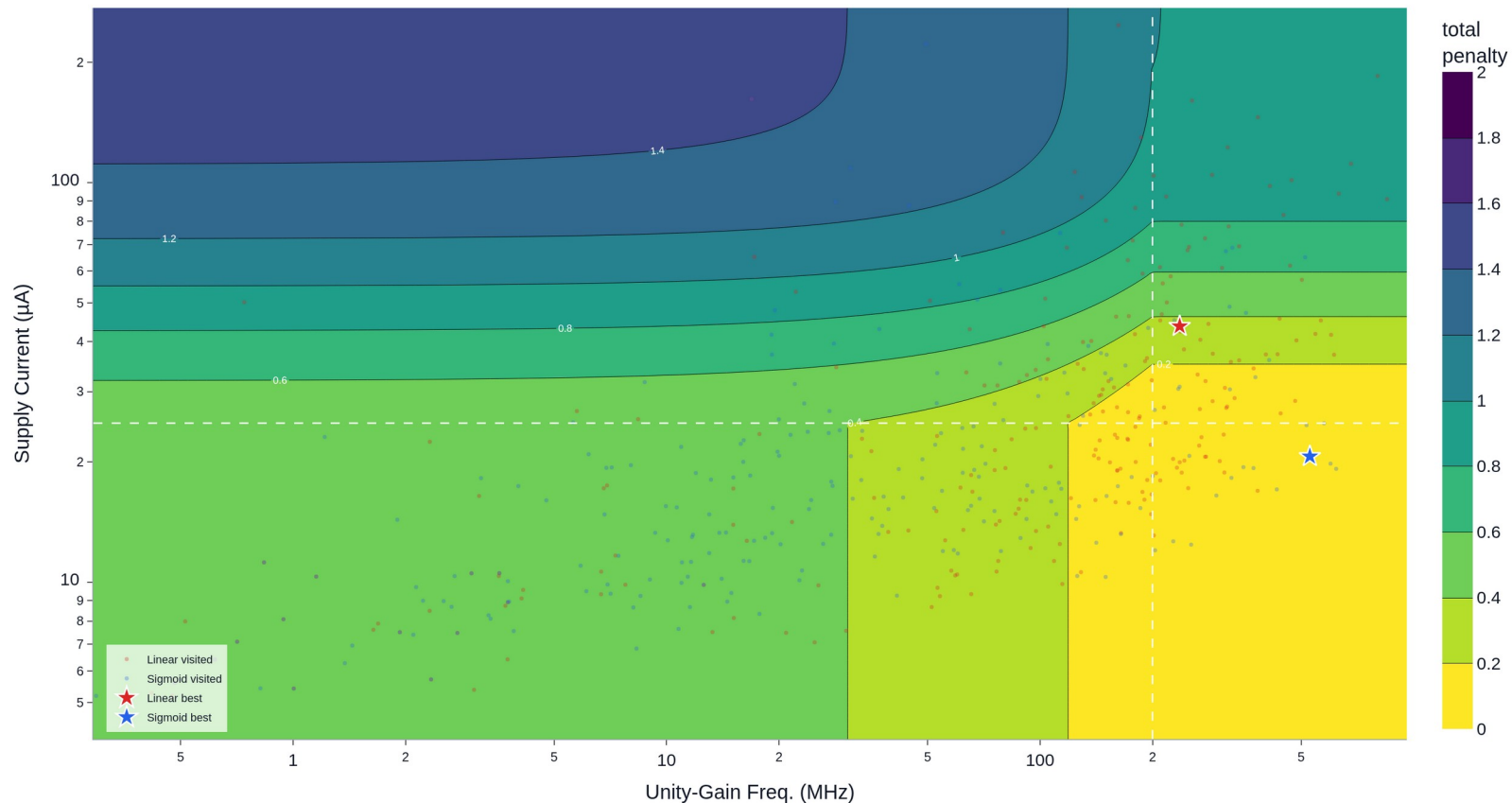
LINEAR BEST

43.7 μA

1.75× over budget

Why: the equi-score landscape

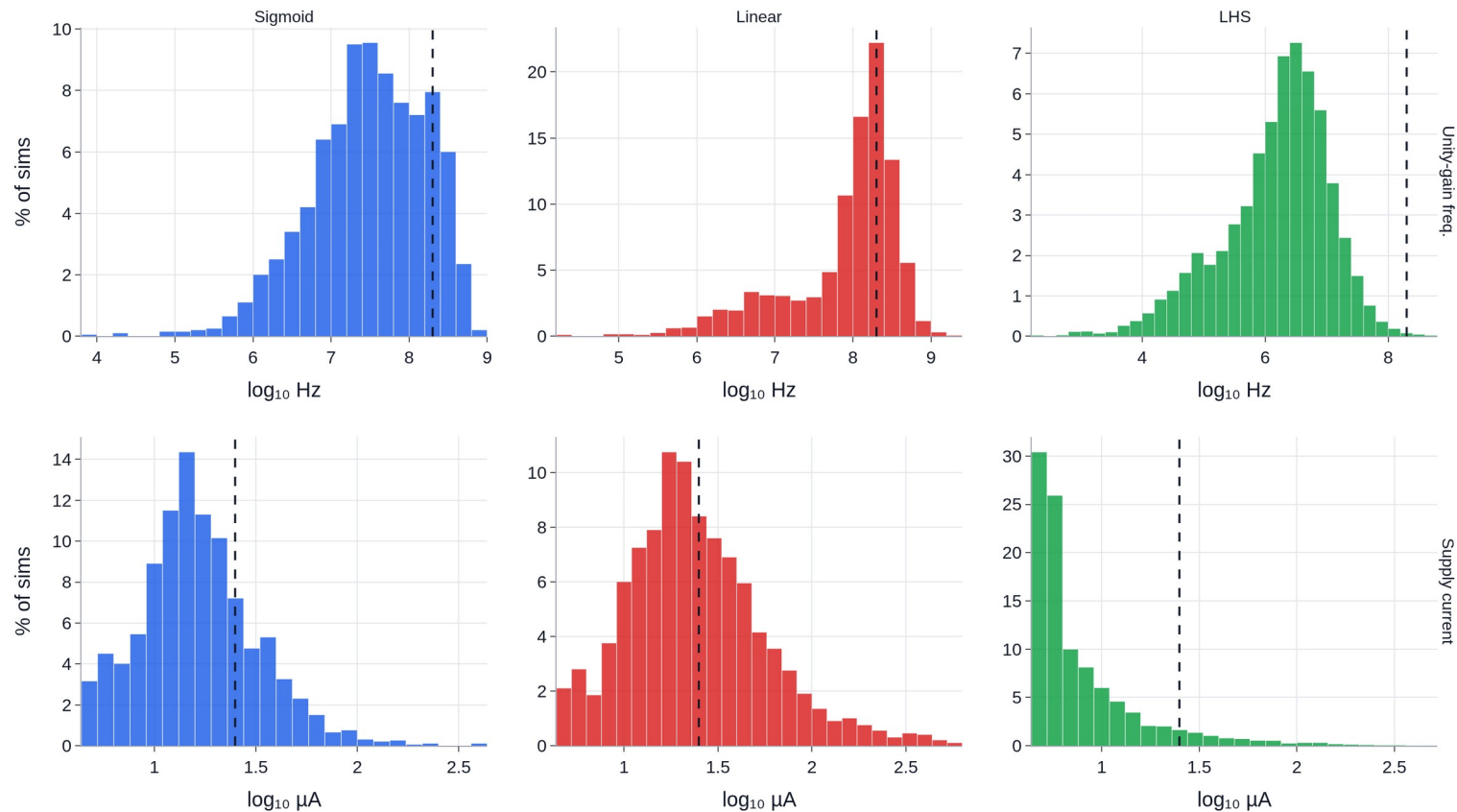
Linear $\rightarrow$ straight, parallel iso-lines dominated by one axis (the other is masked). Sigmoid $\rightarrow$ a bounded, concentric basin pulling toward the feasible corner.



Cost field over bandwidth $\times$ current, with real visited points overlaid

Where each method spends its simulations

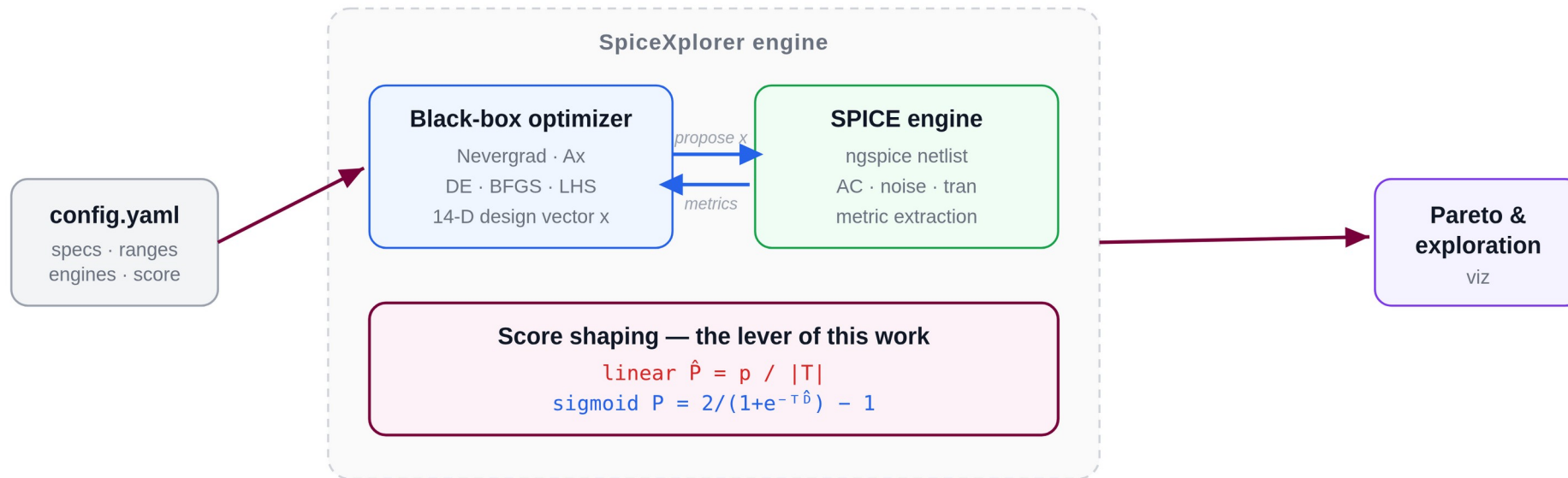
Sigmoid keeps unity-gain-frequency diversity while pulling current *below* 25 μA . Linear over-concentrates at high UGF and higher power; LHS stalls at the $\approx 5 \mu\text{A}$ sub-threshold bias.



Dashed line = spec target

SpiceXplorer — the open framework behind this

Everything you just saw is one YAML config away.



A YAML config orchestrates black-box optimizers (Nevergrad / Ax) against SPICE: metric extraction → score shaping → Pareto / exploration viz.

github.com/MacAnalog/SpiceXplorer • open-source

Agentic, topology-aware sizing

01

Warm starts

Initialize from circuit theory (gm/ID) instead of cold random search.

02

Constraint synthesis

Auto-derive symmetry and biasing constraints to shrink the feasible search.

03

Manager agents

Multi-fidelity orchestration: cheap screens first, full SPICE where it counts.

Formulation is a first-class lever

CONSTRAINT SATISFACTION

Sigmoid met all 6 specs
Linear met 0 / 2000

POWER

20.6 μA vs 43.7 μA
 $\sim 2\times$ lower

SAME SOLVER & BUDGET

Only the score shape changed

Thank you — questions?